Project: Worked examples
Title: Fault protection: Zs, disconnection, adiabatic, fault level
Status: DRAFT Author: TempestOS (2026-10-23) Checker: — (—) Issued: —

1. Scope and method

Automatic disconnection of supply in a TN system for one circuit: the earth fault loop impedance at the conductor's operating temperature against the device's maximum, the disconnection time, the protective conductor (and line conductor) against the adiabatic minimum, and the prospective three-phase fault at the origin against the device's breaking capacity.
Loop impedance: $Z_s = Z_e + (R_1 + R_2)\cdot[1 + \alpha(\theta - 20)]$, $R = \rho_{20} L/S$ Fault current: $I_f = C_{min} U_0/Z_s$, $C_{min} = 0.95$ Maximum loop impedance: $Z_{s,max} = C_{min} U_0/I_a$; I_a = 5, 10 or 20·I_n for MCB types B, C, D, or read from the time–current points at t_{req} (log-log) Adiabatic: $S_{min} = \sqrt{(I^2t)/k}$, $k = \sqrt{[Q_c(\beta + 20)/\rho_{20} \cdot \ln(1 + (\theta_f - \theta_i)/(\beta + \theta_i))]}$ (derived); below 0.1 s the device's let-through I^2t Fault level (simplified): $Z_T = u_k U_n^2/S_r$, $I_{k3} = c U_n/(\sqrt{3} Z_T)$, $U_n = \sqrt{3} U_0$
Ref: BS 7671:2018 (as amended) reg. 411.3.2.2, 411.3.2.3, 411.4.4 — Automatic disconnection in TN systems: times and the loop impedance condition
Ref: BS 7671:2018 (as amended) reg. 543.1.3, 434.5.2 — Protective conductor and fault current adiabatic checks
Ref: BS EN 60204-1:2018 cl. 6.3.3 — Protection by automatic disconnection of supply
Ref: IEC 60909-0:2016 cl. 5.3 — Voltage factor c

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
U_0	Nominal voltage to earth	230	V	spec: Supply characteristics	50 … 1000 V
Z_e	External earth fault loop impedance	0.8	ohm	spec: DNO declared maximum for TN-C-S	0.01 … 3 ohm
C_min	Voltage factor C_min	0.95		assumption: Minimum voltage factor.	0.9 … 1
c_max	Voltage factor c for the maximum fault	1.1		assumption: Maximum voltage factor.	1 … 1.1
L	Circuit length	40	m	drawing: E-LAY-002 rev B	0.1 … 1000 m
S_line	Line conductor cross-section	4	mm^2	drawing: E-SCH-002 rev B	0.5 … 630 mm^2
S_cpc	Protective conductor cross-section	2.5	mm^2	drawing: E-SCH-002 rev B	0.5 … 630 mm^2
theta	Conductor temperature for Z_s	70	degC	assumption: Thermoplastic insulation at full load.	20 … 90 degC
rho_20	Resistivity at 20 °C	0.000017241	ohm*mm	assumption: Annealed copper resistivity.	0.000001 … 0.0001 ohm*mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
alpha	Temperature coefficient of resistance	0.00393	1/K	assumption: Copper temperature coefficient.	0.001 … 0.01 1/K
Q_c	Volumetric heat capacity	0.00345	J/(K*mm^3)	assumption: Copper volumetric heat capacity.	0.001 … 0.01 J/(K*mm^3)
beta	Reciprocal temperature coefficient β	234.5	K	assumption: Copper reciprocal temperature coefficient.	100 … 400 K
theta_i	Initial temperature for the adiabatic check	70	degC	assumption: Thermoplastic operating temperature.	20 … 250 degC
theta_f	Final temperature for the adiabatic check	160	degC	assumption: Thermoplastic limiting temperature.	100 … 400 degC
device	Protective device	MCB type C (I_α = 10 I_n)		drawing: E-SCH-002 rev B	any
I_n	Rated current	16	A	drawing: E-SCH-002 rev B	1 … 1600 A
t_req	Required disconnection time	0.4	s	spec: Final circuit up to 32 A, TN system	0.1 … 5 s
I2t_let	Device let-through energy below 0.1 s	1500	A^2*s	assumption: Read from the MCB let-through chart at the fault current.	1 … 1000000000 A^2*s

3. Assumptions

A1. TN system, a single earth fault of negligible impedance; no motor contribution, arc fault or IT system. (Scope of the check.)
A2. Line and protective conductors are the same material at the same temperature; R = ρ_{20}·L/S corrected by α. (Conductor data from a material record.)
A3. k is derived from the material constants and labelled derived; it is not read from a licensed table. (§A5.)
A4. Fuse and MCCB characteristics are points the user reads off the manufacturer's curve (log-log interpolation; no extrapolation). (No time–current table ships.)
A5. Optional field-test value 0.8 × Z_s,max is a rule of thumb for measurements at ambient temperature. (Labelled on the sheet.)
A6. C_min (Voltage factor C_min) is taken as 0.95. (Minimum voltage factor.)
A7. c_max (Voltage factor c for the maximum fault) is taken as 1.1. (Maximum voltage factor.)
A8. theta (Conductor temperature for Z_s) is taken as 70 degC. (Thermoplastic insulation at full load.)
A9. rho_20 (Resistivity at 20 °C) is taken as 0.000017241 ohm*mm. (Annealed copper resistivity.)
A10. alpha (Temperature coefficient of resistance) is taken as 0.00393 1/K. (Copper temperature coefficient.)
A11. Q_c (Volumetric heat capacity) is taken as 0.00345 J/(K*mm^3). (Copper volumetric heat capacity.)
A12. beta (Reciprocal temperature coefficient β) is taken as 234.5 K. (Copper reciprocal temperature

coefficient.)
A13. theta_i (Initial temperature for the adiabatic check) is taken as 70 degC. (Thermoplastic operating temperature.)
A14. theta_f (Final temperature for the adiabatic check) is taken as 160 degC. (Thermoplastic limiting temperature.)
A15. I2t_let (Device let-through energy below 0.1 s) is taken as 1500 A^2*s. (Read from the MCB let-through chart at the fault current.)

4. Working

[18] R_12 = rho_20 * L * (1 / S_line + 1 / S_cpc) * (1 + alpha * (theta - 20 degC))
= 0.000017241 ohm·mm × 40 m × (1 / 4 mm² + 1 / 2.5 mm²) × (1 + 0.00393 1/K × (70 degC - 20 de
= 0.53635 ohm
R_1 + R_2 at the conductor temperature

[19] Z_s = Z_e + R_12 — BS 7671 reg. 411.4.4
= 0.8 ohm + 0.53635 ohm
= 1.3364 ohm
Earth fault loop impedance

[20] I_f = C_min * U_0 / Z_s
= 0.95 × 230 V / 1.3364 ohm
= 163.51 A
Earth fault current

[21] I_a = 10 * I_n — BS EN 60898-1
= 10 × 16 A
= 160 A
Instantaneous tripping current (10 I_n)

[22] Z_s_max = C_min * U_0 / I_a — BS 7671 reg. 411.4.4
= 0.95 × 230 V / 160 A
= 1.3656 ohm
Maximum earth fault loop impedance

[23] u_Zs = Z_s / Z_s_max
= 1.3364 ohm / 1.3656 ohm
= 0.97856
Loop impedance utilisation

[24] Z_test = 0.8 * Z_s_max
= 0.8 × 1.3656 ohm
= 1.0925 ohm
Field-test maximum at ambient (0.8 × Z_s,max, rule of thumb)

[25] t_d = instantaneous region (I_f ≥ I_a): t ≤ 0.1 s
= I_f = 163.51 A
= 0.1 s
Disconnection time

[27] I2t = device let-through I²t
= 1500 A²s
= 1500 A²·s
Fault energy (let-through below 0.1 s)

[28] k = √[Q_c(β + 20)/ρ₂₀ · ln(1 + (θ_f − θ_i)/(β + θ_i))] (derived) — BS 7671 reg. 543.1.3
= √[0.00345 × 254.5 / 0.000017241 × ln(1 + 90/304.5)]
= 114.84
Adiabatic factor k, A·√s/mm² (derived, compare with the licensed table)

[29] S_min = √(I²t)/k — BS 7671 reg. 543.1.3, 434.5.2
= √1500 / 114.84
= 0.33726 mm²
Minimum conductor cross-section (adiabatic)

[30] u_cpc = S_min / S_cpc
= 0.33726 mm² / 2.5 mm²
= 0.1349
Protective conductor utilisation
[31] u_line = S_min / S_line
= 0.33726 mm² / 4 mm²
= 0.084315
Line conductor utilisation (same fault energy)

5. Results

SYMBOL	VALUE	DESCRIPTION
R_12	0.53635 ohm	$R_1 + R_2$
Z_s	1.3364 ohm	Loop impedance
I_f	163.51 A	Earth fault current
I_a	160 A	Disconnection current
t_d	0.1 s	Disconnection time
Z_s_max	1.3656 ohm	Maximum loop impedance
u_Zs	0.97856	Loop impedance utilisation
Z_test	1.0925 ohm	Field-test maximum (rule of thumb)
k	114.84	Adiabatic k (derived)
S_min	0.33726 mm²	Minimum cross-section
u_cpc	0.1349	Protective conductor utilisation
u_line	0.084315	Line conductor utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Loop impedance within maximum (disconnection in time)	u_Zs = 0.97856	u_Zs ≤ 1	0.979	BS 7671 reg. 411.4.4
PASS	Protective conductor against adiabatic minimum	u_cpc = 0.1349	u_cpc ≤ 1	0.135	BS 7671 reg. 543.1.3
PASS	Line conductor against adiabatic minimum	u_line = 0.084315	u_line ≤ 1	0.0843	BS 7671 reg. 434.5.2
PASS	Disconnection time within required	t_d = 0.1 s	t_req = 0.4 s	0	BS 7671 reg. 411.3.2.2

7. Applicability

WARNING — record.rho_20: rho_20: a tabulated value not taken from a reference record (Conductor material constants); enter it as one so its source, edition and licence travel with it.
WARNING — record.alpha: alpha: a tabulated value not taken from a reference record (Conductor material constants); enter it as one so its source, edition and licence travel with it.
WARNING — record.Q_c: Q_c: a tabulated value not taken from a reference record (Conductor material constants); enter it as one so its source, edition and licence travel with it.
WARNING — record.beta: beta: a tabulated value not taken from a reference record (Conductor material constants); enter it as one so its source, edition and licence travel with it.

WARNING — record.theta_i: theta_i: a tabulated value not taken from a reference record (Conductor material constants); enter it as one so its source, edition and licence travel with it.
WARNING — record.theta_f: theta_f: a tabulated value not taken from a reference record (Conductor material constants); enter it as one so its source, edition and licence travel with it.
WARNING — record.I2t_let: I2t_let: a tabulated value not taken from a reference record (Device time–current points); enter it as one so its source, edition and licence travel with it.

8. Verification

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Template elec.fault-protection@1.0.0 Content hash 18def3b903eb98c6 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0
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